## 1. PROJECT TITLE

# Real-Time Pulse Feed Mobile Application with Notifications and Offline Support

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## 2. ABSTRACT

This project presents the design and development of a real-time mobile application built using Flutter and Firebase. The application provides a unified "Pulse Feed" that displays live updates from multiple sources such as Firebase Firestore, local SMS data, and external APIs like weather services.

The system ensures real-time synchronization, offline data accessibility, and push notifications using Firebase Cloud Messaging (FCM). The app demonstrates modern mobile development concepts such as real-time streaming, cloud integration, offline-first architecture, and multi-source data aggregation.

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## 3. INTRODUCTION

In today's digital world, users require instant access to dynamic information from multiple sources. Traditional applications require manual refresh and lack real-time interaction.

This project introduces a **Pulse Feed system**, which aggregates multiple real-time data streams into a single unified interface. The application continuously listens to updates and displays them instantly without user intervention.

The system is built using Flutter for cross-platform development and Firebase for backend services.

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## 4. PRESENT SYSTEM

The existing systems have several limitations:

- No real-time data updates (manual refresh required)
- Separate applications for different data sources (messages, notifications, weather)
- Limited offline functionality
- Lack of unified user experience
- No integration of multiple data streams

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## 5. PROPOSED SYSTEM

The proposed system introduces a **Real-Time Pulse Feed Application** with the following improvements:

- Unified feed combining multiple data sources
- Real-time updates using Firebase Firestore
- Offline data storage using Hive
- Push notifications using Firebase Cloud Messaging
- Integration of SMS, Firebase data, and Weather API
- Dynamic UI with modern design

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## 6. SYSTEM ARCHITECTURE

### High-Level Architecture

Flutter App (UI)

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Service Layer (Pulse Service)

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| Firestore | SMS | Weather API |

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Local Storage (Hive)

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Notifications (FCM)

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## 7. DATA FLOW ARCHITECTURE

### 🔹 Authentication Flow

- User registers using Firebase Authentication

- Unique User ID (UID) is generated
- User profile stored in Firestore

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🔹 **Real-Time Messaging Flow**

- User sends data → stored in Firestore
- Firestore streams update instantly
- UI updates using StreamBuilder

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🔹 **Multi-Source Feed Flow**

- Firestore data (messages)
- SMS data (device messages)
- Weather API (live updates)
- All merged into a single Pulse Feed

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🔹 **Notification Flow**

- FCM token generated
- Stored in user profile
- Notifications sent via Firebase

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🔹 **Offline Flow**

- Data cached locally using Hive
- Accessible without internet
- Syncs automatically when online

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## 8. FEATURES

### Real-Time Updates

- Instant data synchronization
- No manual refresh required

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**Unified Pulse Feed**

- Combines multiple data sources
- Single scrolling interface

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**Offline Support**

- Uses Hive for local storage
- Data available without internet

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**Push Notifications**

- Firebase Cloud Messaging integration
- Works in foreground & background

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**User Authentication**

- Secure login & registration
- Firebase Authentication

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**Dynamic UI**

- Modern card-based design
- Smooth animations

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## 9. MODULES

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### 🔹 1. Authentication Module

- Handles login and registration
- Firebase Auth integration

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### 🔹 2. Pulse Feed Module

- Displays unified real-time feed
- Combines multiple data streams

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🔹 **3. Messaging Module**

- Private chat system
- Real-time message updates

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🔹 **4. Notification Module**

- Handles push notifications
- FCM integration

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🔹 **5. Offline Module**

- Stores data locally
- Syncs when internet returns

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🔹 **6. Service Layer (Pulse Service)**

- Merges Firestore, SMS, and API data
- Provides unified stream

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**10. TECHNOLOGIES USED**

| Technology | Purpose |
|---|---|
| Flutter | Mobile App Development |
| Dart | Programming Language |
| Firebase Firestore | Real-time Database |
| Firebase Auth | Authentication |
| Firebase Cloud Messaging | Notifications |

| Technology | Purpose |
| --- | --- |
| Hive | Offline Storage |
| connectivity_plus | Network Detection |
| REST API | Weather Data |

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**11. ADVANTAGES**

- Real-time data synchronization
- Unified multi-source data display
- Offline-first architecture
- Scalable cloud-based system
- Improved user experience

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**12. LIMITATIONS**

- Requires internet for real-time sync
- SMS integration depends on permissions
- Notifications need proper setup
- Not optimized for large-scale production

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**13. FUTURE ENHANCEMENTS**

- Add AI-based recommendations
- Improve UI with advanced animations
- Add user personalization
- Integrate more APIs (news, alerts)
- Deploy on Play Store

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**14. CONCLUSION**

The Real-Time Pulse Feed Application successfully demonstrates a modern mobile architecture that integrates real-time data streaming, offline functionality, and multi-source aggregation.

The project provides a scalable foundation for building advanced real-time systems and showcases the power of Flutter and Firebase in developing responsive and efficient mobile applications.

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## 15. REFERENCES


- https://flutter.dev
- https://firebase.google.com
- https://pub.dev
- https://developer.android.com


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